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EXPERIMENT 2

2) Perform Basic Image Handling and Processing Operations on an Image – Read an Image in Python and Convert an Image to Blur using Gaussian Blur

AIM:

To read an image using Python and convert the input image into a blurred image using the Gaussian Blur technique in OpenCV.

Procedure:

1. Install Python and OpenCV (cv2) in the system.
2. Import the OpenCV library in the Python program.
3. Read the input image using the cv2.imread() function.
4. Check whether the image is read successfully.
5. Apply Gaussian Blur using cv2.GaussianBlur() with a suitable kernel size such as (15, 15).
6. Display the original image and the blurred image using cv2.imshow().
7. Save the blurred image using cv2.imwrite(), if required.
8. Close all image windows using cv2.destroyAllWindows().

Sample Input:

Input image: sample.jpg

The program reads the image file sample.jpg from the specified folder.

Sample Python Code:

```
import cv2
```

```
image = cv2.imread('sample.jpg')
```

```
if image is None:
```

```
    print('Image not found')
```

```
else:
```

```
    blurred = cv2.GaussianBlur(image, (15, 15), 0)
```

```
    cv2.imwrite('blurred_image.jpg', blurred)
```

```
    cv2.imshow('Original Image', image)
```

```
    cv2.imshow('Gaussian Blurred Image', blurred)
```

```
    cv2.waitKey(0)
```

```
    cv2.destroyAllWindows()
```

Sample Output:



Result:

Thus, the input image was successfully read using Python and OpenCV, and Gaussian Blur was applied successfully to produce a blurred image.